

Auteur: Victor Pena
Studierichting: Informatica
Oefeningenreeks 1C nr. 48.6

$$z^6 = -64 = -64 + 0i \Rightarrow \begin{cases} x = -64 \\ y = 0 \end{cases}$$

$$r = \sqrt{(-64)^2 + 0^2} = 64$$

$$y = 0, x < 0 \Rightarrow \theta = \pi$$

$$\begin{aligned} w_0 &= \sqrt[6]{64} \operatorname{cis} \left(\frac{\pi + (0)2\pi}{6} \right) \\ &= 2 \operatorname{cis} \left(\frac{\pi}{6} \right) \\ &= 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \\ &= 2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) \\ &= \sqrt{3} + i \end{aligned}$$

$$\begin{aligned} w_1 &= \sqrt[6]{64} \operatorname{cis} \left(\frac{\pi + (1)2\pi}{6} \right) \\ &= 2 \operatorname{cis} \left(\frac{\pi}{2} \right) \\ &= 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \\ &= 0 + 2i \\ &= 2i \end{aligned}$$

$$\begin{aligned} w_2 &= \sqrt[6]{64} \operatorname{cis} \left(\frac{\pi + (2)2\pi}{6} \right) \\ &= 2 \operatorname{cis} \left(\frac{5\pi}{6} \right) \\ &= \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \\ &= -\sqrt{3} + i \end{aligned}$$

$$\begin{aligned} w_3 &= \sqrt[6]{64} \operatorname{cis} \left(\frac{\pi + (3)2\pi}{6} \right) \\ &= 2 \operatorname{cis} \left(\frac{7\pi}{6} \right) \\ &= 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) \\ &= -\sqrt{3} - i \end{aligned}$$

$$\begin{aligned}
w_4 &= \sqrt[6]{64} \operatorname{cis} \left(\frac{\pi + (4)2\pi}{6} \right) \\
&= 2 \operatorname{cis} \left(\frac{3\pi}{2} \right) \\
&= 2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) \\
&= 0 - 2i \\
&= -2i
\end{aligned}$$

$$\begin{aligned}
w_5 &= \sqrt[6]{64} \operatorname{cis} \left(\frac{\pi + (5)2\pi}{6} \right) \\
&= 2 \operatorname{cis} \left(\frac{11\pi}{6} \right) \\
&= 2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) \\
&= \sqrt{3} - i
\end{aligned}$$

References